

Take home notes for:

Ginger bug

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Base Recipe 1: Ginger Bug

Ginger Bug soda is probably my favourite fermented soft drink (favourite fermented drink has to go to beer, sorry everything else). Unlike store bought soda's it's lower in sugar, high in probiotics plus it's flavoured and carbonated naturally. Fizzy pop that's healthy! Yes please.

The Ginger Bug or Starter

A "ginger bug" is a starter, that can be used to make ginger beer or any number of other flavoured sodas, in the same way that people make a sourdough starter to then make sourdough bread.

It captures the natural yeasts and bacteria that live on the skin of fresh ginger and builds them up into an active culture, a form of SCOBY (Symbiotic Community of Yeast and Bacteria). Once established, you add a small amount of your ginger bug to a sugary flavoured liquid, such as a ginger infusion in the case of ginger beer, and ferment it.

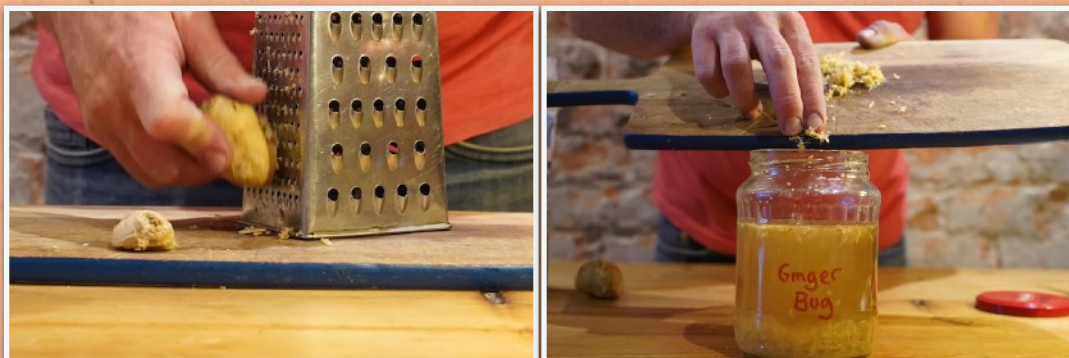
To make a ginger bug starter active you have to feed it daily with ginger and sugar for about a week but once established it can be kept alive indefinitely with less regular feeds, much like a sourdough starter.

Ingredients

- Fresh ginger, unpeeled (organic where you can, it's more likely to have the microbes we need)
- Sugar
- Water

Method

Finely grate or chop 2 tablespoons of ginger (skin on) and add it to a clean jar with 2 tablespoons of sugar and half fill with water. Stir well and cover loosely with a cloth or a screw on lid not fully tightened.



Each day after that, add a tablespoon of grated ginger and a tablespoon of sugar, stirring well each time. Keep the jar at room temperature.

After 3-5 days you should start to see small bubbles rising when you stir, this is a sign the wild yeast and bacteria are multiplying. Keep feeding daily until



it's reliably fizzy within a few hours of feeding, usually 5-7 days in total. If you're having trouble getting it going see **troubleshooting**.

Once active you don't need to keep adding ginger you can just feed it sugar.

Your starter is now ready to use to make a probiotic ginger beer or other kind of natural soda.

Making Natural Sodas

Ingredients

- Ginger Bug Starter
- Brown Sugar or Demerara (white is fine in a pinch)
- Water
- Things to flavour (See **Flavouring Ideas**)

Equipment

- A big clean jar or vessel, at least 1 litre capacity (primary fermentation vessel)
- A cloth or lid to cover
- A sieve or strainer
- Weighing scales
- A Kettle
- Pyrex jug
- Bottles (for the second ferment)

Method

To start with work out how much soda you want to make. This is probably dictated by your fermentation vessel size. The vessel that I usually use (A big kilner jar) holds 2 litres comfortably so I'll go with the amounts for that but will also use percentages, so you can work it out for your fermenting vessel.

Boil a kettle. In a pyrex jug weigh out 6% of the final 2 litres of soda in sugar, so here 120g sugar is 6% on 2 litres. Add to this enough boiling water to dissolve it but not too much, for 2 litres I use 500 ml. Stir until fully dissolved. We now need to dilute this boiled sugar water with cold water, if we

mix boiling water directly with the ginger bug, it will die, so we need to bring the water temp down first.

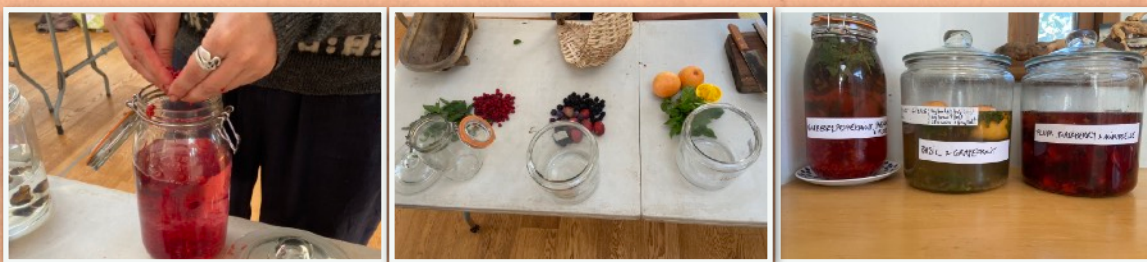
I do this by putting 1.5l cold water into the fermenting vessel, which cools down the hot sugar liquid we pour in next. Add your sugar water and if needed top up to the full 2 litres or your total volume.

...To make ginger beer

This is essentially the same as above with the extra step of infusing ginger pieces like a tea.

In a pan add 500ml water (25% of total) and 120g sugar (6% weight of total water) and bring to a simmer. Add a handful of chopped ginger, you can experiment with the exact amount given how fiery you like it. Simmer for 15 mins then add the rest of the water to cool down, then proceed as below.

Now that you have your sugary liquid, if you want to add flavourings you can add them directly to the fermenting vessel. This is the chance to get creative, See **Flavouring ideas** for more inspiration.



Take your active ginger bug starter, strain off some of the liquid into your fermentation vessel, leaving the ginger pulp behind in the starter jar to keep feeding with water and sugar as described previously.

Place a loose fitting lid or cloth over your fermenting vessel and you're good to leave it on the side to ferment.

First Fermentation

After a day or so you should start to notice small bubbles and rising to the surface, if you've added fruit that might start floating to the top. After about 2 days or when all the fruit is floating it's ready for secondary fermentation. Remove the fruit, which should be floating at the top and strain the liquid into bottles.

Second Fermentation

Once in your bottles, seal them and leave at room temperature for 1-2 days. It will become fizzy in the bottle (See **Why are the drinks fizzy?**). Then move to the fridge over night before consuming.

Fermentation Considerations

Flavouring Ideas

Simply add these things to your primary fermentation.

- **Strawberry, Basil and black pepper** - This might sound mad but it's delightful
- **Apple juice and cinnamon** - Apple juice has enough sugar that you don't need to add your 6% sugar, it's very lively though so watch out for carbonation.
- **Mixed berries and mint** - Frozen berries work great because the freezing breaks the cell walls in the berries and releases more juice when they thaw
- **Elderflower cordial** - Make up a fairly sweet squash and use this in place of the sugar water.

Really try anything you can think of though. My next experiment is Scotts pine needles. Fingers crossed.

Sugar and fruits

In my experience natural sodas are more happy when there are more nutrients. Brown and dark sugar will boost fermentation vigour as will the addition of fruits. If you're finding your soda's aren't fizzy enough you could try adding fruit or changing to a less refined sugar.

Looking after your ginger bug

If you need a break from feeding your ginger bug every few days, you can slow it right down by storing it straight after feeding in the fridge for up to a month. You'll need to bring it back to life after though with a couple of feeds

Why not kombucha?

Within the world of home fermented tonic beverages, Kombucha currently seems to be king. I have made kombucha a lot over the years but these days I only make it for others for the simple fact that as it ages it can become incredibly sour, almost like drinking vinegar and I'm just not a fan of that as a drink. There is tons of information out there if you're interested and I would highly recommend you giving it a try if that's your bag, the process is very similar to making ginger bug drinks so the knowledge is transferable.

Why are the drinks Fizzy?

Ginger bug is a kind of SCOBY a symbiotic community of bacteria and yeast. These SCOBY microbes work collaboratively to consume the sugars and turn them into carbon-dioxide (Co₂) which is what makes the drink fizzy. Whilst consuming the sugars the SCOBY also produces various acids, making the drink lightly tangy and trace amounts of alcohol, which with these sugar levels is about the same as non alcoholic beer.

Temperature

Ginger bug can be very temperature dependent and especially in our colder cumbrian climate, in some rooms they can be very sluggish to get going. Ideally you would want to keep them above 20°C. An airing cupboard or warm part of the house can work or even a very low power greenhouse heated seed mat.

Why a second fermentation?

The first fermentation is mostly about building up a healthy population of microbes and developing flavour. Because that stage happens in a big jar covered with a cloth or loose fitting lid, most of the CO₂ produced simply escapes into the air rather than carbonating the liquid.

Sealing the drink into a bottle for a second ferment traps that Co₂ instead of letting it escape, so it has nowhere to go but into the liquid, this is what gives ginger bug sodas their fizz. Once enough Co₂ has built up, moving the drink to the fridge helps the Co₂ dissolve into the liquid, making the drink retain more fizz.

A note on bottling

WARNING - It is very important that you do not bottle ginger bug drinks too early and then forget about them. Bottling early means more sugar in the bottle, which means more co₂ produced, which means more pressure built up. If this pressure exceeds the bottles pressure rating it could explode and send small pieces of glass flying through the air, not good.

You should never use bottles for olive oil or wine as these are not pressure rated. If you are going to use glass, use bottles designed to hold pressure.

Whilst I don't like using plastic in fermentation where possible, using plastic bottles when you start is a great way to see pressure building, as the bottle literally starts to bulge, rather than exploding.

If you want to use glass but are concerned my advice would be to put some of your batch into a plastic bottle as a tester and the rest in glass bottles. Then as they ferment you can feel the plastic bottle to gage how much pressure is building in the glass bottles.

Troubleshooting

Ginger bug not bubbling after a week

Try a different ginger, an organic one or sometimes even just from a different super market. Some gingers are treated before sale to kill off microbes, which is exactly what we don't want.

A cooler kitchen will also slow things down considerably, try moving the jar somewhere warmer.

Second ferment bottles not carbonating

Usually means not enough residual sugar or yeast activity, or the room is too cold. Add a touch more sugar or fruit juice before sealing, and give it a warmer spot.

Second ferment bottles overflowing when opened

Too much sugar in the solution before bottling leading to over carbonation. Let your first fermentation have a little more time before bottling.